

Balanced mixed determinants of prime cyclic covers: a revised fusion argument

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Abstract

We isolate a proposed consequence of the prime balanced-tuple theorem and Schoen's simple-tuple cycle theorem. For an arbitrary finite list of positive-genus cyclic covers of the projective line of the same odd prime degree, balance of the combined branch residues implies algebraicity of a specified rational space of mixed determinant classes. The argument constructs a compact-type degeneration, uses algebraic smoothing and specialization, and treats disconnected pairing graphs by an explicit rational correspondence. This is a classical proof draft using stated external results, not a machine-checked theorem. No assertion about the full Hodge ring, a Mumford–Tate group, or novelty follows without additional work; the all-powers application is stated separately with its missing input.

1 Scope and conventions

Fix an odd prime p , a primitive root $\zeta = \exp(2\pi i/p)$, and $K = \mathbf{Q}(\zeta)$. Let $N \geq 1$. For each $1 \leq j \leq N$, let

$$f_j : Y_j \longrightarrow \mathbf{P}_{\mathbf{C}}^1$$

be a smooth connected cyclic cover of degree p , with a specified faithful generator σ_j of its deck group. Its ordered nonzero branch residues are $a_{j1}, \dots, a_{js_j} \in \mathbf{F}_p^\times$, where $s_j \geq 3$ and $\sum_i a_{ji} = 0$ in $\mathbf{F}_p$. The residue a means that positive local monodromy is σ_j^a . Set

$$A_j = J(Y_j), \quad V_j = H^1(A_j, \mathbf{Q}) = H^1(Y_j, \mathbf{Q}), \quad h_j = s_j - 2, \quad S = \sum_j s_j, \quad H = \sum_j h_j = S - 2N.$$

We use the induced action on cohomology for the chosen K -action, sending ζ to σ_j^* . For example the corresponding automorphism of $\text{Pic}^0(Y_j)$ can be taken to be pullback by σ_j^{-1} , so that the Abel–Jacobi identification on H^1 has exactly this convention.

The genus is $g_j = (p-1)h_j/2$. Moreover V_j is a K -vector space of rank h_j : its deck-invariant part is $H^1(\mathbf{P}^1, \mathbf{Q}) = 0$, and $\mathbf{Q}[C_p] = \mathbf{Q} \times K$. All h_j are positive. Repetitions among the curves and different choices of generators on the same curve are permitted.

Write $[a]_p \in \{1, \dots, p-1\}$ for the representative of a nonzero residue. The balance condition is

$$2 \sum_{j,i} [ba_{ji}]_p = pS \quad \text{for every } b \in \mathbf{F}_p^\times. \tag{1}$$

It implies that S , and hence H , is even. Since $H > 0$, $H \geq 2$.

For an embedding $\tau_b : K \hookrightarrow \mathbf{C}$ with $\tau_b(\zeta) = \zeta^b$, write $V_{j,b}$ for the associated eigenspace. There is a canonical rational subspace $W \subset H^H(\prod_j A_j, \mathbf{Q})$ specified after extension of scalars by

$$W_{\mathbf{C}} = \bigoplus_{b \in \mathbf{F}_p^\times} \bigotimes_{j=1}^N \bigwedge^{h_j} V_{j,b}. \quad (2)$$

The embedding is the usual exterior-algebra and Künneth embedding. Each summand has dimension one, and $\dim_{\mathbf{Q}} W = p - 1$. Abstractly,

$$W \simeq \bigwedge_K^H \left(\bigoplus_j V_j \right) \simeq \bigotimes_{K,j} \bigwedge_K^{h_j} V_j.$$

These formulas mean the rational realization (2); they do not mean the full tensor product over $\mathbf{Q}$ of the individual determinant spaces.

Theorem 1.1 (Standalone balanced fusion statement). *With the hypotheses above, assume (1). Then*

$$W \subset \text{im} \left(\text{cl} : \text{CH}^{H/2} \left(\prod_j A_j \right)_{\mathbf{Q}} \longrightarrow H^H \left(\prod_j A_j, \mathbf{Q} \right) \right).$$

In particular $W(H/2)$ is a rational Hodge structure of type $(0,0)$ generated by classes of algebraic cycles. No genericity of the branch points, full monodromy, or endomorphism-algebra hypothesis is required.

Status. The following proof reduces this statement to the precise published inputs in Section 3. The project-specific rational projector, residue conversion, gluing, and reduction to those inputs are supplied below. This draft has not received independent mathematical referee review; finite checks and the repository's Lean layer do not certify its algebraic geometry.

2 Rational realization and algebraic projectors

We first make the tensor product over K usable on algebraic cycles. Let $U_1, \dots, U_m$ be copies of the V_j , with their specified deck operators T_r . On $\bigotimes_{\mathbf{Q},r} U_r$, set

$$e_{uv} = \frac{1}{p} \sum_{a=0}^{p-1} T_u^a T_v^{-a}, \quad E_m = \prod_{r=2}^m e_{1r}, \quad E_1 = \text{id}. \quad (3)$$

All operators in these formulas act on their indicated tensor slots.

Lemma 2.1. *The rational operator E_m is an idempotent with image $\bigoplus_b \bigotimes_r U_{r,b}$. It is induced by an algebraic correspondence on the product containing the tensor slots. For m identical slots, it commutes with every ordinary tensor permutation.*

Proof. Over a splitting field, the value of e_{uv} on an eigentensor with labels b_u, b_v is

$$\frac{1}{p} \sum_{a=0}^{p-1} \zeta^{a(b_u - b_v)} = \begin{cases} 1 & b_u = b_v, \\ 0 & b_u \neq b_v. \end{cases}$$

This proves the image and idempotence. Equality of all labels is invariant under permutations, proving the commutation assertion after scalar extension and hence over $\mathbf{Q}$. Each T_r^a comes from a deck automorphism on an abelian factor. Their graphs, rational linear combinations, and compositions are algebraic correspondences. The H^1 Künneth projection is also algebraic: for an abelian variety of dimension g and an integer $n \geq 2$, the polynomial

$$\pi_k = \prod_{0 \leq d \leq 2g, d \neq k} \frac{[n]^* - n^d}{n^k - n^d}$$

acts as the projection to H^k . It is represented cohomologically by a rational algebraic correspondence. Only this cohomological assertion about π_k is needed, not an equality of Chow motives. $\square$

For an abelian variety A and $m \geq 1$, let $\mu : A^m \rightarrow A$ be addition, $\delta : A \rightarrow A^m$ the diagonal, and $P = \pi_1^{\otimes m}$. The map

$$\iota_m = P\mu^* : H^m(A, \mathbf{Q}) \longrightarrow H^1(A, \mathbf{Q})^{\otimes m}$$

is the unnormalized alternating inclusion. The cup product $q_m = \delta^*|_{H^1(A)^{\otimes m}}$ satisfies

$$q_m \iota_m = m! \text{ id}, \quad \iota_m q_m = m! \text{ Alt}_m, \quad \text{Alt}_m = \frac{1}{m!} \sum_{\pi \in \mathfrak{S}_m} \text{sgn}(\pi) P_\pi. \quad (4)$$

Indeed expansion of $\mu^*v = \sum_r \text{pr}_r^* v$ for $v \in H^1(A)$ gives the first identity; the exterior-algebra relation gives the second. Geometric permutations on products of odd cohomology classes have Koszul signs. Thus P_π here is the ordinary tensor permutation, realized by the geometric permutation correspondence multiplied by $\text{sgn}(\pi)$. There is no hidden assumption that these two actions coincide.

For $A = A_j$ and $m = h_j$, the correspondence

$$Q_j = \frac{1}{h_j!} q_{h_j} E_{h_j} \iota_{h_j} \quad (5)$$

is an idempotent on $H^{h_j}(A_j)$ with image $D_j = \bigoplus_b \bigwedge^{h_j} V_{j,b}$. Idempotence follows from (4) and Lemma 2.1. This proves rationality and algebraic realizability of the determinant summands without selecting a single complex embedding as a rational subspace.

Remark 2.2 (The scalar action must be induced from one slot). On D_j , the scalar action of $k \in K$ is the action obtained from k on *one* slot after applying E_{h_j} and alternation. All slot actions then agree, so this defines a unital $\mathbf{Q}$ -algebra action. The corresponding scalar ζ acts by ζ^b on the b determinant line. In contrast, $\bigwedge^{h_j}(\sigma_j^*)$ acts by ζ^{bh_j} . These are different operators, and the latter is trivial when $p \mid h_j$. One must not use this exterior-power deck action to identify embeddings between determinant factors. Our construction needs no hypothesis $\gcd(p, h_j) = 1$.

Apply E_H to all raw H^1 slots simultaneously, alternate within the j -th block of h_j slots, and use (4) to return to $H^H(\prod_j A_j)$. This gives a rational algebraic projector Q_W with image W . More explicitly, put $\iota = \bigotimes_j \iota_{h_j}$, $q = \bigotimes_j q_{h_j}$, and $c = \prod_j h_j!$. On the indicated Künneth degree $(h_1, \dots, h_N)$ the formula is

$$Q_W = c^{-1} q E_H \iota; \quad (6)$$

extend it by zero on the other degrees using algebraic Künneth projectors. The same calculation proves idempotence. Its action takes algebraic classes to algebraic classes and fixes W .

3 Exact external inputs

The proof uses the following results; none is asserted as a new theorem here.

1. **Prime balanced tuples.** A tuple of nonzero residues of even length at least four, with sum zero modulo the odd prime p , satisfying (1), can be partitioned into pairs $(a, -a)$. This is the prime equality $B_p^n = D_p^n$ stated on p. 24 before Theorem A of Aoki [1], where the prime case is credited to W. Parry; here $n = S - 2$ is positive and even. The 1984 erratum corrects the introduction’s Theorem B, not this input.
2. **Simple-tuple cycles.** For a smooth connected cyclic curve with faithful action and simple branch tuple (a union of opposite pairs), Schoen’s Theorem 2.0 and Corollary 3.1 [2] give algebraic classes spanning the primitive determinant space on its primitive Jacobian factor. We use this in characteristic zero, where Theorem 2.0 also holds over an arbitrary algebraically closed field. For a prime cover of $\mathbf{P}^1$ the primitive factor is the whole Jacobian up to isogeny, as shown in Section 1. The transfer in Corollary 3.1 uses the algebraicity of inverse Lefschetz on abelian varieties; its argument works over algebraically closed characteristic-zero fields as well, by descent and comparison. No claim about arbitrary abelian varieties of Weil type is used.
3. **Balanced admissible-cover deformation.** Abramovich–Corti–Vistoli [4], Theorems 3.0.2 and 4.3.2 and Corollary 3.0.5, identify balanced admissible G -covers with balanced twisted covers and give their unobstructed deformation theory, with a flat map to the moduli stack of stable pointed target curves. Section 3 identifies the balanced node smoothing directions.
4. **Compact-type Picard and specialization.** For a proper nodal curve family whose fibers have tree dual graph, Pic^0 is an abelian scheme, with special fiber the product of the component Jacobians. The relative Picard input is the semistable Picard theorem attributed to Deligne in [5, §9.4, Theorem 1, p. 259]; the special-fiber description is [5, §9.2, Example 8, pp. 246–247]. An explicitly inspected primary use of precisely this relative assertion is Achter–Pries [6, §3.1]. For a smooth proper scheme over a DVR, specialization of Chow classes is compatible with smooth-proper specialization in ℓ -adic cohomology. For the cycle map see SGA 6, X.7.13, especially 7.13.11, and X.7.16, [7, pp. 576–579, 585]. Section 7.13 assumes a henselian trait; we make this base change explicitly in Section 5. Schoen [2, pp. 30–31] invokes the same compatibility. The cycle-level construction and base change are recorded in [8, Tags 0H4H, 0H4M].

Source-audit boundary. Aoki’s exact prime assertion and erratum were pinned by the earlier repository audit. In this repair the indicated Schoen statements, his specialization paragraph, the 1998 addendum, ACV’s deformation argument and equivalence, the Achter–Pries Picard assertion, and the Stacks cycle construction were inspected. The final cross-review directly inspected the original BLR pages 246–247 and 258–259, and the original SGA 6 pages 576–579 and 585. The latter verify the henselian assumption, smooth-proper comparison, rational cycle-class compatibility in 7.13.11, and passage to geometric fibers in 7.16. We do not claim to have inspected the scan’s missing sections 7.14–7.15. These are standard imported results with the precise forms used here stated above, not new theorems proved by this manuscript.

4 From balance to a compact-type cover

Choose a pairing of all S branch occurrences into opposite residues using the first input. Form a multigraph Γ with vertices $1, \dots, N$ and one edge for every chosen pair; both ends can lie on the same vertex. An occurrence is used once. Let I be a connected component of this graph. If $|I| > 1$, choose a spanning tree T_I from its non-loop edges; if $|I| = 1$, use the empty tree. Every branch occurrence on a vertex of I is paired with another occurrence in I .

Lemma 4.1 (Residue versus tangent character). *At a totally ramified branch point with residue a , the action of σ_j on a local tangent coordinate is multiplication by $\zeta^{a^{-1}}$. Consequently residues a and $-a$ have inverse tangent characters for the chosen common deck generator.*

Proof. Write the cover locally as $x = u^p$. Positive monodromy about $x = 0$ sends u to ζu . By definition this is σ_j^a , so if $\sigma_j(u) = \zeta^c u$ then $ac = 1$ modulo p . Also $(-a)^{-1} = -a^{-1}$. $\square$

Lemma 4.2 (Balanced gluing). *For every I there is a connected balanced admissible C_p -cover $Y_{I,0} \rightarrow B_{I,0}$ whose component covers are exactly the chosen $Y_j \rightarrow \mathbf{P}^1$ for $j \in I$, whose source and target dual graphs are T_I , and whose remaining branch tuple is simple. Its source has compact type and genus $\sum_{j \in I} g_j$.*

Proof. For each tree edge glue its two indicated base branch points, and glue their unique ramification points upstairs. The deck action agrees under gluing, since the glued points are fixed. The curve maps glue to a finite morphism of nodal curves; away from markings and nodes they are principal C_p -bundles. Lemma 4.1 gives the balanced action at every node. Both graphs are the chosen tree because a totally ramified point has a unique preimage.

Each target component has exactly s_j special points: replacing a selected branch marking by a node changes none of their number. Hence the pointed target is stable. Each source component has positive genus. A genus-one component in a non-singleton tree has a node and is stable; a singleton graph component has even $s_j \geq 4$, hence $g_j \geq 2$. Thus the source is stable after forgetting branch markings as well. The cover satisfies the admissible G -cover definition of [4, Definition 4.3.1].

Only complete pairs belonging to T_I were removed. The unremoved tuple is therefore a union of opposite pairs and has length

$$s_I = \sum_{j \in I} s_j - 2(|I| - 1) = H_I + 2, \quad H_I = \sum_{j \in I} h_j. \quad (7)$$

All original occurrences within I are paired, so their total number is even; thus H_I is positive and even, $H_I \geq 2$, and $s_I \geq 4$. The tree graph has first Betti number zero, giving both compact type and the asserted genus. $\square$

Lemma 4.3 (Algebraic simultaneous smoothing). *The cover of Lemma 4.2 is the closed fiber of an algebraic family of admissible C_p -covers over the spectrum of a DVR essentially of finite type over $\mathbf{C}$. Its geometric generic fiber is a smooth connected cyclic cover of $\mathbf{P}^1$ with the simple remaining branch tuple. After a finite base change, the family has an abelian scheme $\mathcal{A}_I = \text{Pic}^0(\mathcal{Y}_I/R)$, equipped with the chosen K -action, whose special fiber is $\prod_{j \in I} A_j$.*

Proof. At a glued node of residue a the local smoothing is

$$uv = t, \quad x = u^p, \quad y = v^p, \quad xy = t^p, \quad \sigma(u, v) = (\zeta^{a^{-1}}u, \zeta^{-a^{-1}}v).$$

This is equivariant and smooths the node when $t \neq 0$. Notice the power t^p in the target parameter. The cover map is finite; it is not asserted flat at a node. Both nodal curve families are flat over the parameter base.

By the ACV input these local balanced deformations globalize in an algebraic stack with smooth local charts. More precisely the local-to-global deformation sequence for a nodal twisted curve surjects onto its local node deformation spaces: the obstruction to this surjection is H^2 of its tangent sheaf, which is zero for a curve; the node deformation sheaf is supported at finitely many points. The invariant smoothing line survives exactly because the tangent characters are inverse. The torsor adds no obstruction, as its target BG has zero cotangent complex in characteristic zero. These are the deformation calculations in ACV Section 3. Thus no node parameter is forced to vanish.

Take an algebraic smooth atlas at our point and a curve through it with tangent direction outside the finitely many node-parameter hyperplanes. Such a curve exists: choose a suitable line in local étale coordinates and take its inverse image. Normalize and localize at the point, obtaining a DVR. Each source node parameter has a simple zero, so the source total space is regular at these nodes. All node parameters are nonzero at the generic point. Shrinking the atlas if necessary, smoothness away from the specified nodes is open, so the generic source and target are smooth. This constructs an algebraic family with the prescribed closed fiber, not merely a formal or analytic plumbing. The empty tree case can instead use a constant family over a DVR.

The source is geometrically connected: its connected special fiber has $h^0(\mathcal{O}) = 1$, which bounds the generic value from above by semicontinuity, while constants bound it from below. The target has genus zero and the remaining marked sections; after a finite extension if needed it is $\mathbf{P}^1$. The labeled inertia data at these sections is locally constant, so the generic tuple is exactly that in (7).

There are only generic and closed geometric fibers over the DVR; both have compact type. The compact-type Picard input gives the abelian scheme. For completeness, on the closed fiber normalization gives a Picard extension with torus of rank $b_1(T_I) = 0$. Restriction of line bundles therefore identifies its identity component with the product of the component Jacobians, compatibly with deck actions. Since $1 + \sigma + \cdots + \sigma^{p-1}$ acts as zero on the generic Jacobian, it acts as zero on the abelian scheme: a homomorphism of separated group schemes equal to zero on the dense generic fiber is zero. Thus the deck action factors through K in rational endomorphisms everywhere. $\square$

5 Spreading cycles and specializing their classes

Fix I , let L be the fraction field of the DVR in Lemma 4.3, and fix an algebraic closure $\bar{L}$. The geometric generic curve has K -rank H_I in H^1 , by (7). The simple-tuple theorem gives cycle classes spanning

$$W_{I, \bar{\eta}} = \bigwedge_{K \otimes_{\mathbf{Q}} \mathbf{Q}_{\ell}}^{H_I} H^1(\mathcal{A}_{I, \bar{\eta}}, \mathbf{Q}_{\ell}) \subset H^{H_I}(\mathcal{A}_{I, \bar{\eta}}, \mathbf{Q}_{\ell}).$$

Here and below the wedge notation denotes the matching-embedding realization after extension to $\mathbf{Q}_{\ell}$, including the appropriate Tate twist when comparing cycle classes. Fix any rational prime ℓ .

Lemma 5.1. *The space*

$$W_I = \bigoplus_b \bigotimes_{j \in I}^{h_j} V_{j, b} \subset H^{H_I} \left(\prod_{j \in I} A_j, \mathbf{Q} \right)$$

is generated by algebraic classes.

Proof. Choose finitely many rational algebraic cycles whose $\mathbf{Q}_{\ell}$ -classes span the geometric generic determinant space, applying the generic-fiber algebraic determinant projector first if the supplied

cycles have other components. Each cycle is defined over a finite extension of L : its finitely many defining equations have coefficients algebraic over L . A single finite extension L'/L suffices for all chosen cycles. The DVR is excellent, so its integral closure in L' is finite; localize at a point above the closed point to obtain a DVR R' . Its residue field is $\mathbf{C}$. Pulling back the abelian scheme leaves its complex special fiber unchanged.

Take the closures of the generic cycles in this proper smooth scheme and specialize in the Chow group. An integral closure of a generic cycle dominates the DVR and is flat over it; its special fiber with multiplicities gives the specialization. For the cohomological comparison, now pass from R' to its strict henselization R'^{sh} , with a compatible choice of geometric generic point. Its residue field is still $\mathbf{C}$, so its closed fiber and the specialized complex cycles are unchanged. Pull back the abelian scheme and the cycles to this trait. Specialization commutes with this flat DVR base change; all closures and finite-extension descent were already constructed over R' . The henselian hypothesis in SGA 6, X.7.13 is now met. The compatibility cited in Section 3 identifies these classes with the original generic classes under the smooth-proper ℓ -adic specialization isomorphism. This is a specialization of cycles defined over a finite extension, not an assumption that an arbitrary Hodge class is algebraic because it lies in a family.

The relative H^1 local system carries the deck action. The projectors from Section 2, formed from addition, diagonals, multiplication, and deck maps of the abelian scheme, commute with specialization. Thus specialization identifies the generic determinant space with $\bigwedge_K^{H_I}(\bigoplus_{j \in I} V_j)$ on the closed fiber. Expanding the top exterior power of each b -eigenspace gives exactly W_I : to reach degree $H_I = \sum h_j$ one must use all h_j dimensions of each summand. In particular no other Künneth degrees occur.

Apply the algebraic special-fiber projector if necessary so that every specialized cycle class lies in W_I . Their rational Betti classes have $\mathbf{Q}_\ell$ -span $W_I \otimes \mathbf{Q}_\ell$. Betti-étale comparison and extension of scalars then show that their $\mathbf{Q}$ -span is W_I : the quotient of W_I by that span becomes zero over the field $\mathbf{Q}_\ell$, and is therefore zero. This proves rational, not merely ℓ -adic, algebraicity. $\square$

Proof of Theorem 1.1. For every connected component I of the pairing graph, Lemma 5.1 supplies algebraic classes spanning W_I . Take their external products. They span $\bigotimes_{\mathbf{Q}_\ell, I} W_I$, which contains independent embedding choices on the different graph components. The projector Q_W of (6) kills precisely the terms whose embedding choices disagree. Its image on that tensor product is W , since every matching term is fixed. Applying this algebraic correspondence to the external-product cycles proves the theorem. If the graph is connected, this last projection is redundant; if it is disconnected, it is essential. $\square$

6 Independent Hodge-type check and example

The Hodge type can also be checked without a monodromy theorem. In a coordinate with infinity unramified write $y^p = \prod_i (x - t_i)^{a_i}$ with $1 \leq a_i \leq p - 1$ and $\sum a_i = kp$. For $l = p - b$, forms of deck eigenvalue ζ^b have the shape $r(x)dx/y^l$. Holomorphy at t_i forces $r(x)$ divisible by $(x - t_i)^{\lfloor la_i/p \rfloor}$, since the valuation is $p \text{ord}_{t_i}(r) + p - 1 - la_i$. Holomorphy at infinity bounds the remaining polynomial degree by $lk - \sum_i \lfloor la_i/p \rfloor - 2$. Consequently

$$\dim V_{j,b}^{1,0} = -1 + \frac{1}{p} \sum_i [-ba_{ji}]_p = s_j - 1 - \frac{1}{p} \sum_i [ba_{ji}]_p.$$

The fractional-part sum is a positive integer, so the formula includes the zero-dimensional case. Under balance, the sum over j is $S - N - S/2 = H/2$. Each line of (2) therefore has type $(H/2, H/2)$. This also fixes the sign convention rather than importing an ambiguous Chevalley–Weil formula.

For $p = 5$, take branch tuples

$$(1, 1, 3), \quad (1, 2, 4, 4, 4).$$

They give K -ranks 1, 3 and genera 2, 6. Their $H^{1,0}$ multiplicities for $b = 1, 2, 3, 4$ are respectively $(1, 1, 0, 0)$ and $(1, 1, 2, 2)$, whose sum is $(2, 2, 2, 2)$. The concatenation pairs into three $(1, 4)$ pairs and one $(3, 2)$ pair. Glue the residue-3 point of the first curve to the residue-2 point of the second. The remaining tuple is $(1, 1, 1, 4, 4, 4)$, the smooth fused curve has genus 8 and K -rank 4, and the theorem gives a four-dimensional rational space of codimension-two classes on $A_1 \times A_2$. Neither individual determinant space is of middle Hodge type. The mixed balance is the relevant condition.

7 The separate all-powers implication

Let A be an abelian variety assembled, up to isogeny, from cyclic quotient Jacobians of odd prime degree. To deduce the Hodge conjecture for every power of A , the following *additional generator statement* suffices:

Generator hypothesis. Every rational Hodge class on every A^m is in the span obtained from divisor classes and spaces W satisfying (1) by external products, cup products, pullbacks and push-forwards along algebraic correspondences, including the isogenies, quotient maps, permutations, and rational projectors needed for realization. The determinant words have positive multiplicities of the chosen cyclic H^1 factors; conjugate words use inverse chosen generators. Any dualization is implemented with its polarization and Tate twist before using this form.

Corollary 7.1 (Conditional all-powers application). *Assume the generator hypothesis for A . Then every rational Hodge class on every power of A is algebraic.*

Proof. Divisor Hodge classes are algebraic by the Lefschetz $(1, 1)$ theorem. Theorem 1.1 handles each indicated determinant space. Every operation in the generator hypothesis preserves algebraic cycle classes. $\square$

The generator hypothesis is a distinct representation-theoretic and rational descent problem. This manuscript does not prove the claimed full monodromy, the treatment of compact embedding sectors, graph-arrow signs, the complete central torus, or exhaustion by determinant words. An integer lattice basis of determinant relations alone does not prove generation of the effective tensor monoid; negative exponents need separately realized duals and twists, or an appropriate monoid-generation argument. No unconditional all-powers claim is made here.

8 Relation to prior work and review gates

The input for a single simple cyclic cover is Schoen's. Compact-type clutching and algebraic specialization are also established tools; the proposed contribution is their combination for an arbitrary list of separately unbalanced prime cyclic factors whose combined signatures balance, including the disconnected-graph projection. A bounded literature search did not locate this exact quantified mixed-product formulation, but does not establish novelty. It should be compared explicitly with the cyclic-cover clutching literature and later work on exceptional classes on products of CM and moving factors.

Schoen's 1998 addendum [3] corrects the missing polarization invariant in the statement of the 1988 Theorem 3.2. It explicitly continues to use Corollary 3.1. Our argument uses the latter simple-tuple input and does not import the uncorrected general fourfold statement.

The next review gates are: independent expert verification of the geometric proof beyond the internal cross-review; an exact prior-art comparison for the mixed-product statement; and, for any global Hodge application, a proof of the separate generator hypothesis. The finite script accompanying this revision checks projector identities and tuple/graph examples only. It cannot verify smoothability, cycle existence, or specialization.

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